

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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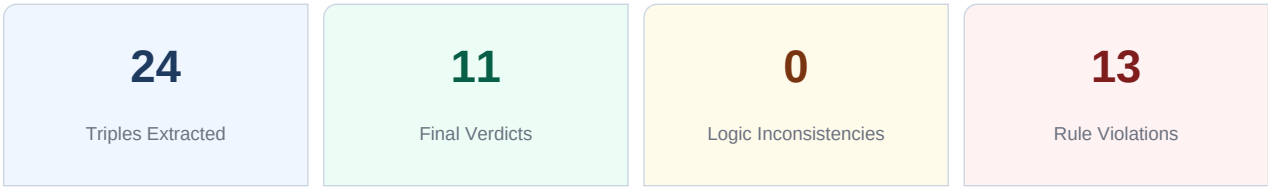
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REPORT OVERVIEW & METHODOLOGY

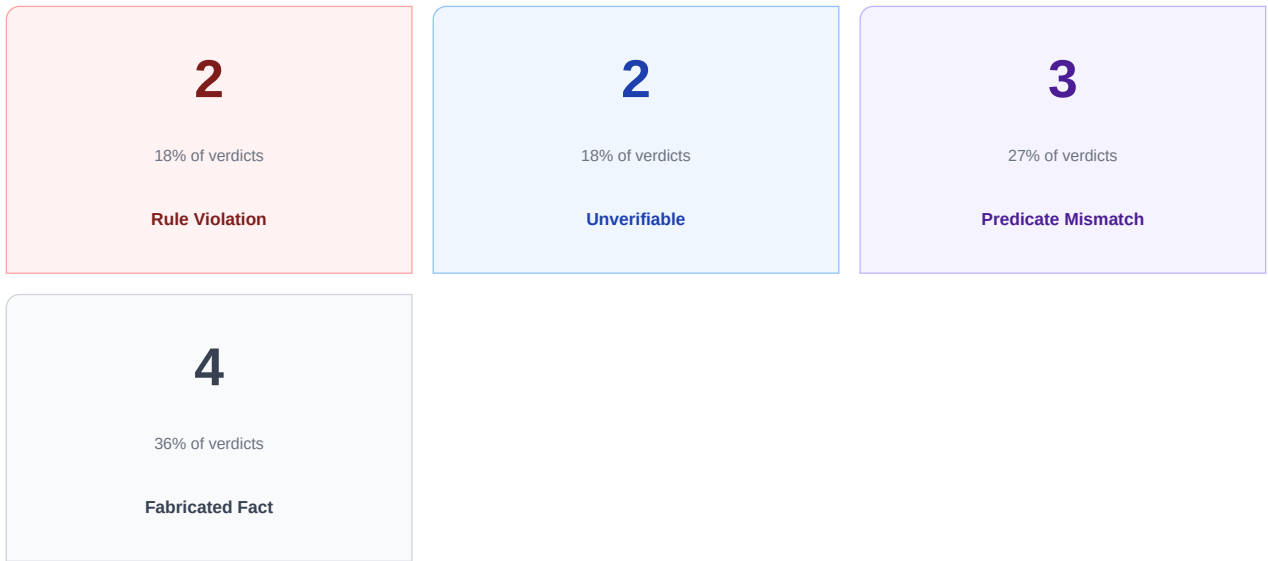
This document outlines the results of the KG-TripleGuard verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	890	HAS_INTEREST_RATE	30		The interest rate associated with the subject.
2	290	HAS_INTEREST_RATE	20		The interest rate associated with the subject.
3	333	HAS_INTEREST_RATE	40		The interest rate associated with the subject.
4	333	HAS_RISK_TIER	high prime		The risk tier classification assigned to the subject.
5	598	HAS_RISK_TIER	high prime		The risk tier classification assigned to the subject.
6	633	HAS_RISK_TIER	high prime		The risk tier classification assigned to the subject.
7	729	HAS_RISK_TIER	high prime		The risk tier classification assigned to the subject.
8	833	HAS_RISK_TIER	high prime		The risk tier classification assigned to the subject.
9	890	HAS_RISK_TIER	high prime		The risk tier classification assigned to the subject.
10	290	HAS_RISK_TIER	high prime		The risk tier classification assigned to the subject.
11	301	HAS_RISK_TIER	deep subprime		The risk tier classification assigned to the subject.
12	333	HAS_APPROVAL_ODDS	200		The likelihood or odds of approval for the subject.
13	290	HAS_APPROVAL_ODDS	30		The likelihood or odds of approval for the subject.
14	890	HAS_APPROVAL_ODDS	40		The likelihood or odds of approval for the subject.
15	300	HAS_INTEREST_RATE	2		The interest rate associated with the subject.
16	303	HAS_INTEREST_RATE	3		The interest rate associated with the subject.
17	333	HAS_INTEREST_RATE	4		The interest rate associated with the subject.
18	777	HAS_INTEREST_RATE	6		The interest rate associated with the subject.
19	100000	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, assets, or a down payment (e.g., 10-20% of the loan amount), but this varies by lender and loan type.		The required account balance associated with the subject.
20	300	HAS_DESCRIPTION	Deep subprime credit score, indicating very high credit risk. Borrowers in this range face extremely high interest rates, limited loan options, and low approval odds.		A textual description of the subject.
21	300	HAS_APPROVAL_ODDS	1		The likelihood or odds of approval for the subject.
22	305	HAS_APPROVAL_ODDS	2		The likelihood or odds of approval for the subject.

#	Subject	Predicate	Object	Type	Predicate Definition
23	405	HAS_APPROVAL_ODDS	3		The likelihood or odds of approval for the subject.
24	730	HAS_APPROVAL_ODDS	4		The likelihood or odds of approval for the subject.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	HAS_INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
290	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	40	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
598	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
633	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
729	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
833	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
290	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
301	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
333	HAS_APPROVAL_ODDS	200	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_APPROVAL_ODDS	30	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
890	HAS_APPROVAL_ODDS	40	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
300	HAS_INTEREST_RATE	2	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
303	HAS_INTEREST_RATE	3	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	4	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
777	HAS_INTEREST_RATE	6	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM

Subject	Original Predicate	Object	Canonical Predicate	Conf.
100000	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, assets, or a down payment (e.g., 10-20% of the loan amount), but this varies by lender and loan type.	HAS_REQUIRED_ACCOUNT_BALANCE	LOW
300	HAS_DESCRIPTION	Deep subprime credit score, indicating very high credit risk. Borrowers in this range face extremely high interest rates, limited loan options, and low approval odds.	HAS_DESCRIPTION	LOW
300	HAS_APPROVAL_ODDS	1	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
305	HAS_APPROVAL_ODDS	2	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
405	HAS_APPROVAL_ODDS	3	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
730	HAS_APPROVAL_ODDS	4	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM

FACTUALLY INCONSISTENT TRIPLES

All triples labelled `factually_inconsistent` from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	c_interest_rate_range	Credit score 890 is above 850, violating the constraint.
factually_inconsistent	RULE_VIOLATION	290	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	c_interest_rate_range	Credit score 290 is below 300, violating the constraint.
factually_inconsistent	RULE_VIOLATION	333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	40	c_interest_rate_range	Interest rate 40% is above the maximum allowed of 30%, violating the constraint.
factually_inconsistent	RULE_VIOLATION	333	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_300_579	Credit score 333 is within the 300-579 range and object does not contain 'deep subprime'.
factually_inconsistent	RULE_VIOLATION	598	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_580_619	Credit score 598 is within 580–619 and object does not contain 'subprime'.
factually_inconsistent	RULE_VIOLATION	633	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_620_659	Credit score 633 is within 620–659 and object does not contain 'near prime'.
factually_inconsistent	RULE_VIOLATION	729	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	Credit score 729 is within 660–739 and object does not contain 'prime'.
factually_inconsistent	RULE_VIOLATION	833	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_740_850	Credit score 833 is within 740–850 and object does not contain 'super prime'.
factually_inconsistent	RULE_VIOLATION	890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_740_850	Credit score 890 is above 850, exceeding the valid tier range for any prime.
factually_inconsistent	RULE_VIOLATION	290	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_300_579	Credit score 290 is below 300 and does not fit a valid risk tier.
factually_inconsistent	RULE_VIOLATION	333	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	200	c_approval_odds_range	Approval odds 200 is out of 0-100 range, violating constraint.
factually_inconsistent	RULE_VIOLATION	290	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	30	c_approval_odds_range	Credit score 290 is less than 300, violating constraint.

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	40	<i>c_approval_odds_range</i>	Credit score 890 is greater than 850, violating constraint.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
301	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>SEMANANTIC</i>	—	●●●●●● ■■■ 0.60	Predicate and object match some DB entries, but subject '301' is not present. <i>Ev: subject: 375, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 380, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 385, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 390, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 395, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime</i>
300	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	2	Rule Violation	factually_inconsistent	RULE_VIOLATION	<i>STRICT</i>	OBJECT	●●●●●● ●●● 1.00	Subject and predicate match, but object '2' contradicts DB value '22.94'. <i>Ev: subject: 300, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.94</i>
303	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	3	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>NONE</i>	—	●●■■■■■ ■■■ 0.20	Predicate is known but there are no DB entries at all for this triple.
333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	<i>NONE</i>	—	●●■■■■■ ■■■ 0.20	Predicate is known but subject and object do not exist in DB.

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
777	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	6	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	—	●●●●●● ■■■ 0.20	Predicate is in schema but DB contains no supporting evidence for this triple.
100000	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, assets, or a down payment (e.g., 10-20% of the loan amount), but this varies by lender and loan type.	Unverifiable	unverifiable	—	NONE	—	■■■■■■ ■■■ 0.00	Predicate is not known in the DB schema; no matching DB evidence available.
300	HAS_DESCRIPTION	Deep subprime credit score, indicating very high credit risk. Borrowers in this range face extremely high interest rates, limited loan options, and low approval odds.	Unverifiable	unverifiable	—	NONE	—	■■■■■■ ■■■ 0.00	Predicate 'HAS_DESCRIPTION' is not present in the DB schema.
300	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	1	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRICT	OBJECT	●●●●●● ●●● 1.00	Subject and predicate match, but object value '1' contradicts DB value '8'. Ev: subject: 300, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 8

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	2	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	NONE	PREDICATE	●●●●●■ ■■■ 0.60	Subject exists in DB but not with the claimed predicate; only other predicates found. Ev: subject: 305, predicate: RISK_TIE R_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 305, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.61
405	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	3	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	NONE	PREDICATE	●●●●●■ ■■■ 0.60	Subject is present in DB but the predicate does not match any entry for it. Ev: subject: 405, predicate: RISK_TIE R_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 405, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 17.61
730	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	4	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	STRICT	PREDICATE	●●●●●■ ■■■ 0.60	The subject '730' exists, but there is no record with the predicate 'APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD' for this subject in the database. Ev: subject: 730, predicate: RISK_TIE R_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: prime; subject: 730, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 7.29

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.